



COURSE SYLLABUS						
Course Code	COMP 019	Course Title	Applications Development and Emerging Technologies		Course Credit	3
Course Description		This course offers an overview of key emerging technologies like AI, IoT, Blockchain, Cloud Computing, AR/VR, 5G, Cybersecurity, and Quantum Computing, showing how they shape modern application development. Through lectures and hands-on projects, students learn to design and build innovative, real-world solutions while focusing on ethical use and future readiness in software development.				
Pre-Requisites		COMP 009	Co-Requisites	None		
VISION		A Leading Comprehensive Polytechnic University in Asia				
MISSION		Advance an inclusive, equitable, and globally relevant polytechnic education towards national development.				
QUALITY POLICY STATEMENT		The Polytechnic University of the Philippines commits to provide inclusive and equitable quality education and promote lifelong learning opportunities for human advancement and social transformation through re-engineered polytechnic academic programs. Toward this end, we, the members of the PUP Community will vigorously and steadfastly endeavor to continuously improve the standard of University services to the satisfaction of our clients through the adoption and continuous review of our Quality Management System.				
INSTITUTIONAL LEARNING OUTCOMES (ILO)		As a polytechnic state university, PUP shall develop its students to possess: Critical and Creative Thinking. Graduates use their rational and reflective thinking as well as innovative abilities to life situations in order to push boundaries, realize possibilities, and deepen their interdisciplinary, multidisciplinary, and/or transdisciplinary understanding of the world. Effective Communication. Graduates apply the four macro skills in communication (reading, writing, listening, and speaking), through conventional and digital means, and are able to use these skills in solving problems, making decisions, and articulating thoughts when engaging with people in various circumstances. Strong Service Orientation. Graduates exemplify strong commitment to service excellence for the people, the clientele, industry, and other sectors. Adept and Responsible Use or Development of Technology. Graduates demonstrate optimized and responsible use of state-of-the-art technologies of their profession. They possess digital learning abilities, including technical, numerical, and/or technopreneurial skills. Passion for Lifelong Learning. Graduates perform and function in society by taking responsibility in their quest for further improvement through lifelong learning.				





	6. Leadership and Organizational Skills. Graduates assume leadership roles and become leading professionals in their respective disciplines by equipping them with appropriate organizational skills. 7. Personal and Professional Ethics. Graduates manifest integrity and adherence to moral and ethical principles in their personal and professional circumstances. 8. Resilience and Agility. Graduates demonstrate flexibility and the growth mindset to adapt and thrive in the volatile, uncertain, complex and ambiguous (VUCA) environment. 9. National and Global Responsiveness. Graduates exhibit a deep sense of nationalism as it complements the need to live as part of the global community where diversity is respected. They promote and fulfill various advocacies for human and social development.
COLLEGE GOALS AND OBJECTIVES	The Polytechnic University of the Philippines – Taguig Campus aims to achieve the following goals: 1. Offer curricula which are relevant and responsive to the changing needs of the industry and society; 2. Provide trainings to students that will enable them to become potent instruments to socio-economic development. 3. Prepare students to become productive, creative, innovative, and dynamic in their chosen fields of specialization. 4. Provide state-of-the-arts facilities of learning to optimize student development; 5. Inculcate appropriate values that are necessary to build a humane, disciplined, nationalist, and independent society; 6. Tap potentials of students, faculty, administrative staff and other stakeholders in formulating policies for institutional development; 7. Involve all stakeholders in enhancing programs for student’s faculty and employees’ development; 8. Partner with industry and other research institutions strengthening research capabilities of faculty, employees and students; 9. Facilitate presentation of research outputs in international fora, and their publication in recognized journals; 10. Undertake outreach and research-based extension programs by tapping the expertise and other resources of all stakeholders and, 11. Produce technologies for commercialization of livelihood improvement.



PROGRAM LEARNING OUTCOMES (PLO)	Based on CMO	Alignment to ILOs								
		1	2	3	4	5	6	7	8	9
	Apply knowledge of computing, science, and mathematics appropriate to the discipline.									
	Analyze complex problems, and identify and define the computing requirements appropriate to its solution									
	Identify and analyze user needs and take them into account in the selection, creation, evaluation and administration of computer-based systems.									
	Design, implement, and evaluate computer-based systems, processes, components, or programs to meet desired needs and requirements under various constraints.									



	Integrate IT-based solutions into the user environment effectively.									
	Assist in the creation of an effective IT project plan.									
	Apply knowledge through the use of current techniques, skills, tools and practices necessary for the IT profession									
	Understand best practices and standards and their applications.									
	Function effectively as a member or leader of a development team recognizing the different roles within a team to accomplish a common goal.									
	Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.									
	Analyze the local and global impact of computing information technology on individuals, organizations, and society.									
	Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.									
	Recognize the need for and engage in planning self-learning and improving performance as a foundation for continuing professional development									

COURSE LEARNING OUTCOMES (CLO)		Alignment to PLOs							
		1	2	3	4	5	6	7	8
	CLO1. Explain the fundamental concepts, principles, and importance of emerging technologies in application development.								
	CLO2. Identify and analyze current and emerging technology trends and their applications in real-world scenarios.								
	CLO3. Demonstrate understanding of Artificial Intelligence, Machine Learning, Internet of Things, Blockchain, and other emerging technologies.								



	CLO4. Apply cloud-based and edge computing solutions to support scalable and efficient application systems.								
	CLO5. Develop prototypes or applications that integrate at least one emerging technology to solve a specific problem.								
	CLO6. Evaluate the usability, performance, and security implications of applications using emerging technologies.								
	CLO7. Collaborate effectively in teams to design and present technology-driven solutions aligned with industry best practices.								
	CLO8. Reflect on ethical, social, and future implications of adopting emerging technologies in application development								



OUTCOMES-BASED TEACHING AND LEARNING PLAN (OBTL PLAN)

Week	Desired Learning Outcomes (DLOs)	Learning Content/ Topics	Instructional Delivery Design		Assessment		Alignment to CLOs			
			Face-to-Face	Flexible Teaching and Learning Activities (FLTAs)			1	2	3	4
					Face-to-Face	FLTAs				
1	- Equip students on the pivotal ideas embedded in the vision, mission, and 10 Pillars of PUP - Gain an overview of the course subject and requirements	- Orientation about PUP Vision, Mission, and 10 Pillars - Overview about the Subject Course and Course Requirements	Present the VMGO, policies, and guidelines on the conduct of Face-to-Face; and of the requirements of the course.	Read course syllabus and policies Watch: 'The Future of Application Development' video lecture	Formative Assessment (Sharing of Learning)		X			
2	Understand the concept and significance of Emerging Technologies	Definition, characteristics, and importance of emerging technologies; Overview of Top 10 Emerging Trends.	Lecture with examples of current technologies.	Online discussion forum and video presentation.	Short quiz and recitation.	Online quiz and reflection.				
3	Explore Artificial Intelligence (AI) and its applications in app development.	AI Overview, Applications, and Tools in App Development.	Demonstration of AI tools in app prototypes.	Watch AI case study videos and submit insights.	Discussion and quiz.	Identify and describe an AI-powered app.				

 WORLD UNIVERSITY RANKINGS

 2025

7th in the Philippines

92nd in Southeast Asia

 STARS

★★★★★

★★★★★

★★★★



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Bachelor of Science in Information Technology



4	Understand Machine Learning concepts and frameworks.	ML Algorithms, Frameworks (TensorFlow, PyTorch).	Lecture and coding demo.	Coding exercises via online IDEs.	Performance task: simple ML activity.	Submit output of ML exercise.				
5	Discuss Internet of Things (IoT) and its application in systems.	IoT Concepts, Components, Use Cases.	Demonstration using IoT simulation tools.	Online IoT activity: design a system diagram.	Observation checklist.	Submit IoT system plan.				
6	Recognize Blockchain principles and app integration.	Blockchain: Structure, Smart Contracts, and Security Applications.	Lecture and group discussion.	Blockchain case study discussion.	Written quiz.	Submit blockchain use case analysis.				
7	Understand Cloud Computing and its scalability features.	Cloud Computing Models: IaaS, PaaS, SaaS.	Cloud deployment demo.	Cloud lab activity (synchronous).	Quiz and recitation.	Reflection journal submission				
8	- Analyze Edge Computing and its advantages. - Explore AR/VR applications and development.	- Edge vs Cloud Computing; Real-world Applications. - Extended Reality: AR/VR Tools (Unity, ARKit), Use Cases.	- Guided lecture and comparison. - Demo and simulation videos.	- Research and discuss advantages asynchronously. - Online viewing of AR/VR apps in education.	Participation and Q&A	Short essay reflection.				



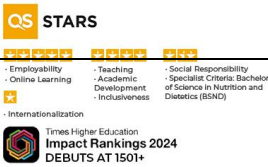
9	MIDTERM WRITTEN EXAMINATION									
10	Examine 5G Technology and its role in real-time applications.	5G Features, Use Cases, Integration in Apps.	Lecture and discussion.	Online simulation or infographic activity.	Quiz and recitation.	Discussion board participation.				
11	Discuss cybersecurity issues and secure development.	Cybersecurity Fundamentals, Data Privacy, Secure Coding.	Lecture with real-world examples.	Online cybersecurity awareness session.	Written test.	Cybersecurity quiz completion.				
12	Understand Quantum Computing and its potential in development.	Quantum Computing: Concepts and Applications.	Visual presentation and discussion.	Watch video, submit summary.	Class discussion.	Essay on quantum computing.				
13	Integrate multiple emerging technologies in app concepts.	Frameworks combining AI, IoT, Cloud, etc.	System design workshop	Collaborative brainstorming (online).	Group evaluation of designs.	Submit integration plan.				
14	Develop and propose app prototype using emerging tech.	Project Proposal Development: Concept, Objectives, Tech Stack.	Prototype proposal defense (face-to-face).	Peer review of draft proposals.	Group evaluation of designs.	Upload proposal file.				
15	Implement prototype using	Application Development and Testing.	Supervised coding lab.	Asynchronous coding and testing.	Code review and observation.	Upload working demo.				



	emerging technologies.									
16	Evaluate prototype performance and scalability.	Prototype Evaluation: Usability, Efficiency, Tech Effectiveness.	Presentation and peer evaluation.	Peer feedback activity.	Rubric-based evaluation.	Submit reflection on feedback.				
17	Present final project integrating emerging technologies.	Final Project Presentation and Demonstration.	Capstone presentation.	Online portfolio documentation.	Final defense assessment.	Submit documentati on.				
18	FINAL WRITTEN EXAMINATION [Indicate what is the final written exam by the professor]									



		Resources and Applying Strategies in Teaching Prose <i>Competency 4:</i> Recognizing Assessment in Teaching Prose	achievement which are consistent with the selected competencies • Learning Plan Making according to the English curricula that develops higher order thinking skills of learners through the use of prose and dramatic texts		• Quizzes					
18	FINAL EXAMINATION									
LIST OF REFERENCES BOOKS: WEBSITE: - Artificial Intelligence (AI) & Machine Learning (ML) IBM. (n.d.). <i>What is artificial intelligence?</i> IBM. https://www.ibm.com/cloud/learn/what-is-artificial-intelligence Google AI. (n.d.). <i>Google AI research.</i> https://ai.google/research/ NVIDIA Developer. (n.d.). <i>AI and deep learning.</i> https://developer.nvidia.com/deep-learning Microsoft. (n.d.). <i>Introduction to machine learning.</i> Microsoft Learn. https://learn.microsoft.com/en-us/training/paths/introduction-machine-learning/ - Internet of Things (IoT) Cisco. (n.d.). <i>What is IoT?</i> Cisco Systems. https://www.cisco.com/c/en/us/solutions/internet-of-things/overview.html IoT For All. (n.d.). <i>Latest trends and case studies.</i> https://www.iotforall.com/										





Oracle. (n.d.). *Internet of Things (IoT) overview*. <https://www.oracle.com/internet-of-things/>

- **Blockchain & Web3**

IBM. (n.d.). *Blockchain technology overview*. <https://www.ibm.com/topics/what-is-blockchain>

101 Blockchains. (n.d.). *Learning hub*. <https://101blockchains.com/>

CryptoZombies. (n.d.). *Learn Solidity by building games*. <https://cryptozombies.io/>

- **Cloud & Edge Computing**

Amazon Web Services. (n.d.). *What is cloud computing?* <https://aws.amazon.com/what-is-cloud-computing/>

Microsoft. (n.d.). *Azure fundamentals*. Microsoft Learn. <https://learn.microsoft.com/en-us/training/modules/fundamental-azure-concepts/>

IBM. (n.d.). *Edge computing explained*. <https://www.ibm.com/topics/edge-computing>

- **Extended Reality (AR/VR)**

Unity Technologies. (n.d.). *Introduction to AR and VR development*. Unity Learn. <https://learn.unity.com/>

Meta for Developers. (n.d.). *AR/VR technology resources*. <https://developers.facebook.com/>

ARPost. (n.d.). *News on AR, VR, and MR*. <https://arpost.co/>

- **5G & Advanced Connectivity**

Ericsson. (n.d.). *The future of 5G*. <https://www.ericsson.com/en/5g>

Huawei. (n.d.). *5G use cases and applications*. <https://carrier.huawei.com/en/spotlight/5g>



- **Cybersecurity**

Cybersecurity and Infrastructure Security Agency (CISA). (n.d.). *Cybersecurity best practices*. <https://www.cisa.gov/topics/cybersecurity-best-practices>

Kaspersky. (n.d.). *Security blog*. <https://www.kaspersky.com/blog/>

IBM. (n.d.). *What is cybersecurity?* <https://www.ibm.com/topics/cybersecurity>

- **Quantum Computing**

IBM. (n.d.). *Quantum learning*. <https://www.ibm.com/quantum>

Microsoft. (n.d.). *Azure Quantum*. <https://azure.microsoft.com/en-us/products/quantum/>

Quanta Magazine. (n.d.). *News and research*. <https://www.quantamagazine.org/>

- **General & Multi-Trend Resources**

Simplilearn. (n.d.). *Top 10 technology trends*. <https://www.simplilearn.com/top-technology-trends-and-jobs-article>

TechNewsWorld. (n.d.). *Emerging tech* section. <https://www.technewsworld.com/section/technology/emerging-tech/>

ISACA. (n.d.). *Emerging technology resources*. <https://www.isaca.org/resources/emerging-technology-resources>

Emerging Tech Brew. (n.d.). *Innovation and trends*. <https://www.emergingtechbrew.com/>

Forbes Technology Council. (2021, September 17). *15 expert-recommended resources for keeping up with the latest tech news*. Forbes.

<https://www.forbes.com/councils/forbestechcouncil/>





CLASSROOM POLICY	
FACE-TO-FACE DELIVERY	FLEXIBLE TEACHING AND LEARNING ACTIVITIES (FLTAs)
Please refer to the FLTA policies, except for the stipulation on Attendance. Attendance will be checked during face-to-face sessions.	Students' Responsibilities Students are expected to: a. Take time to read and watch the provided class learning materials. b. Fulfill class requirements such as posting on discussion boards and submitting group presentations. c. Check the status of their submissions and inform the instructor if they are having problems submitting the requirements.



The instructor may not be able to remind all students about their missing/unsubmitted requirements. If at the end of the course students still have missing requirements, the instructor will compute for the final grade without those submissions (They will be graded with 0) and submit the grade. If the final grade is "INC" grade, the student must contact the instructor to decide in completing the course, ASAP.

d. Confer with classmates or the instructor about the discussions during the synchronous sessions that they missed if unable to attend the live sessions. Recordings will be provided but may not be available immediately.

e. Inform the instructor of concerns that may prevent them from fully participating in the class. This includes withdrawal from the course, long term illnesses, personal matters, and others. Teachers will not be able to help you if you don't tell them of these concerns.

Attendance.
We recognize that students have varying access levels to technology and connectivity, which is why for the most part, the course is conducted asynchronously. During synchronous sessions, there will be no checking of attendance. However, everyone is encouraged to participate in these sessions. A recording and/or transcript of synchronous classes will be made available for students who are unable to attend a synchronous session.

Late Submissions.
Deadlines will be set for the submission of course requirements. This allows the instructor sufficient time to review them. It also helps students pace their work. If you experience problems submitting your work on time, please contact the instructor ahead of time. All requirements must be completed to pass the course. Late submissions will not merit deductions but may not receive timely feedback.

Cheating and Plagiarism.
Any work submitted by a student in this course for academic credit will be the student's own work. Any form of academic dishonesty will be fully dealt with according to the stipulations of the PUP Student Handbook.

Participation.
Participation during discussions is greatly encouraged, and students should feel free to ask questions when an idea or lesson is unclear or confusing, especially during synchronous sessions.

Announcements.
Class announcements will be communicated through the class president. Other communication shall always be coursed through university-issued (i.e., official) email accounts. Students must use their PUP student accounts.

Intellectual Property and Copyright.
Original materials created for this online class is covered by university policies on intellectual property. Some (third-party) instructional materials (e.g., YouTube videos) might be under the public domain or are open access, while others (e.g., some assigned readings) might be subject to fair use. All third-party instructional materials will be attributed to their original creators. Electronic copies of the assigned readings are for your private consumption only. Access to the course materials for this class for PUP students will be restricted to only those students registered in this course and will be available for only





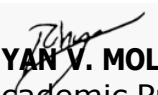
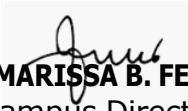
	A limited period. Please do not share these materials with anyone elsewhere. The same applies to the recordings of the synchronous sessions.	
COURSE REQUIREMENT/S	GRADING SYSTEM	
- Developing a Project Proposal on Application Development (focusing on Completeness, Appropriate Use of Emerging Technology, and Innovative Design Process) - Application Prototype Demonstration and Presentation (focusing on Technical Proficiency, System Functionality, User Interface Design, Creativity, and Diligence in Project Implementation.	Mid Term	Final Term
	Class Standing 70% Exercises/Activities Summative Quizzes Recitation and Attendance Reflection Paper	Class Standing 70% Exercises/Activities Summative Quizzes Recitation and Attendance Reflection Paper
	Midterm Examinations 30%	Final Examinations (Teaching Demonstration) 30%
	100%	100%
	Final Grade = (Midterm Grade + Final Term Grade) / 2	
CLASS INFORMATION		FACULTY INFORMATION
Course, Year and Section: BSIT 3 - 1		Name of Faculty: SAN LUIS, ANGELO JOSHUA BANDA
Day and Time: S/S 2:00 PM – 5:00 PM / 5:00 PM – 7:00 PM		Consultation Time:
Room: TGA205 / TGA205		Office Tel. No./ Mobile Phone No.
School Year:	Semester:	E-Mail Address:





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Bachelor of Science in Information Technology



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THE COUNTRY’S 1st POLYTECHNICU

**2025**
7th in the Philippines
92nd in Southeast Asia
541-560 in Asia

**WURI**
The WORLD UNIVERSITY RANKINGS for INNOVATION

Rank 21
GENERATIVE AI APPLICATION

Rank 39
STUDENT SUPPORT ENGAGEMENT

Rank 50
SYMBOL PROMOTION

**STARS**

★★★★★
Employability
Online Learning

★★★★★
Teaching
Academic Development
Inclusiveness

★★★★★
Social Responsibility
Specialist Criteria: Bachelor of Science in Nutrition and Dietetics (BSND)

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